

Jaeyoung Cho

Ph.D. Candidate in Chemistry | Henkelman Group
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EDUCATION

University of Texas at Austin

Austin, TX

Ph.D. in Chemistry

December 2026 (Expected)

Advisor: Graeme Henkelman

Dissertation: Computational Investigation of Ion Diffusion and Phase Transformation Mechanisms in Solid-State Battery Materials Using Machine-Learning Interatomic Potentials

Seoul National University

Seoul, Republic of Korea

M.S. in Materials Science and Engineering

February 2020

Advisor: Jang-Joo Kim

Thesis: Modeling Charge Transport in Organic Hosts: BCP and NBPhen

Yonsei University

Seoul, Republic of Korea

B.S. in Chemistry; B.S. in Materials Science and Engineering

February 2018

RESEARCH INTERESTS

Accelerated Dynamics & Sampling Methods: Machine-learning interatomic potential development for atomistic simulations; long-timescale dynamics (adaptive kinetic Monte Carlo, parallel replica dynamics); transition-state search methods (Dimer, NEB, solid-state NEB).

Materials & Systems for Energy Storage: Design rules for solid-state electrolyte materials and structure–property relationships; phase transformation of battery materials; anode-free battery systems.

RESEARCH EXPERIENCE

Henkelman Research Group, University of Texas at Austin

Austin, TX

Graduate Research Assistant

August 2022 – Present

Solid Electrolyte Interphase Interrelation for Anode-Free Battery Systems

Investigated the role of a lithiophilic interphase layer (Li_2Te) in Li deposition on a copper current collector for stable anode-free all-solid-state lithium batteries. Performed DFT simulations to assess the thermodynamic stability of continuous Li electrodeposition on passivated surfaces via binding-energy comparison.

Halide-Based Electrolytes for All-Solid-State Batteries

Studied entropic effects on the ionic-transport mechanism and screened new cathode materials. Applied quantum-mechanical and accelerated-dynamics methods (adaptive kinetic Monte Carlo, machine-learning molecular dynamics) to quantify anharmonic effects (thermal expansion, internal-energy changes in defect migration) and to resolve sub-lattice ordering effects on diffusion in superionic conductors.

Lawrence Livermore National Laboratory

Livermore, CA

Computational Chemistry & Materials Science Graduate Intern

June 2024 – December 2024

Elucidating Phase Evolution in Boron Nitride through Large-Scale Machine-Learning Molecular Dynamics

Developed and validated a MACE machine-learning potential for boron-nitride polymorphs, then used machine-learning molecular dynamics, enhanced sampling, and solid-state NEB to quantify energy barriers and interfacial evolution pathways between distinct BN phases, advancing understanding of BN polymorphism and phase stability.

Korea Institute of Science and Technology

Full-Time Research Intern

Seoul, Republic of Korea

June 2020 – December 2020

Effect of Organic Ligands on Nanoparticles for Polymer Electrolyte Membrane Fuel Cells (PEMFCs)

Used DFT to study non-stoichiometric CeO_x nanoparticles as radical scavengers, evaluating the thermodynamic favorability and binding energetics of radical species (•OH and •OOH) and the electronic origin of enhanced radical-scavenging activity through density-of-states analysis.

Max Planck Institute for Polymer Research

Visiting Research Student

Mainz, Germany

July 2019 – October 2019

Building a Molecular Library of OLED Host Materials

Performed quantum-mechanical and molecular-dynamics simulations to compute electronic properties of organic molecules and analyze structure–property relationships, identifying critical parameters for high conductivity and contributing to a public molecular database for OLED host materials.

Seoul National University

Graduate Research Assistant

Seoul, Republic of Korea

January 2018 – January 2020

Modeling Charge Transport in Organic Hosts for OLEDs

Combined DFT, molecular dynamics with a polarizable force field, and kinetic Monte Carlo to extract charge-carrier mobility in organic host materials and identify the structural origins of mobility differences across molecular substituents.

HONORS, AWARDS & FELLOWSHIPS

Professional Development Award, Graduate School, University of Texas at Austin December 2025

Merit-Based Scholarship, College of Engineering, Seoul National University July 2019

Merit-Based Scholarship, College of Engineering, Seoul National University March 2018

PUBLICATIONS

Peer-Reviewed Journal Articles

4. Yixian Wang, Vikalp Raj, Kaustubh G. Naik, Bairav S. Vishnugopi, **Jaeyoung Cho**, Mai Nguyen, Elizabeth A. Recker, Yufeng Su, Hugo Celio, Andrei Dolocan, Zachariah A. Page, John Watt, Graeme A. Henkelman, Qingsong Howard Tu, Partha P. Mukherjee, David Mitlin, “Control of Two Solid Electrolyte Interphases at the Negative Electrode of an Anode-Free All-Solid-State Battery Based on Argyrodite Electrolyte,” *Advanced Materials*, 37, 2410948 (2025). [\[link\]](#)
3. Lei Li, Ryan A. Ciufo, Jiyoung Lee, Chuan Zhou, Bo Lin, **Jaeyoung Cho**, Naman Katyal, Graeme Henkelman, “Atom-Centered Machine-Learning Force Field Package,” *Computer Physics Communications*, 292, 108883 (2023). [\[link\]](#)
2. Yixian Wang, Yijie Liu, Mai Nguyen, **Jaeyoung Cho**, Naman Katyal, Hongchang Hao, Ruyi Fang, Nan Wu, Jagjit Nanda, Graeme Henkelman, John Watt, David Mitlin, “Stable Anode-Free All-Solid-State Lithium Battery through Tuned Metal Wetting on the Copper Current Collector,” *Advanced Energy Materials*, 35, 2206762 (2023). [\[link\]](#)
1. Anriban Mondal, Leanne Paterson, **Jaeyoung Cho**, Kun-Han Lin, Bas van der Zee, Gert-Jan A. H. Wetzelaer, Andrei Stankevych, Alexander Vakhnin, Jang-Joo Kim, Andrey Kadashchuk, Paul W. M. Blom, Falk May, Denis Andrienko, “Molecular Library of OLED Host Materials – Evaluating the Multiscale Simulation Workflow,” *Chemical Physics Reviews*, 2, 031304 (2021). [\[link\]](#)

Review & Tutorial Articles

- *[Invited]* Sung Hoon Jung, Ilgar Baghishov, **Jaeyoung Cho**, Akksay Singh, Graeme Henkelman, “A Tutorial Review of Saddle-Point Search Methods for Transition States in Materials and Chemistry,” *Invited; Expected to be submitted to The Journal of Physical Chemistry*, August 2026.

- Kit Barker, **Jaeyoung Cho**, Graeme A. Henkelman, Ieuan D. Seymour, “Halide Materials as Solid Ion Conductors: A Review,” *In preparation; expected submission to Journal of Materials Chemistry A, September 2026.*

Manuscripts in Preparation & Under Review

- Jon A. Newnham, **Jaeyoung Cho**, Tiannou Jin, John W. Still, Graeme Henkelman, Ieuan D. Seymour, “Induced Structural Disorder Enables High Ionic Conductivity and Capacity in $\text{Li}_2\text{Fe}_{1-x}\text{Mn}_x\text{Cl}_4$ Cathodes,” *Chemistry of Materials. Under review. Preprint: ChemRxiv, [\[link\]](#)*
- **Jaeyoung Cho**, Ieuan D. Seymour, Graeme Henkelman, “Mechanistic Studies of Li-Ion Transport in Halide Solid-State Electrolytes via Machine-Learning-Accelerated Adaptive Kinetic Monte Carlo,” *In preparation; Expected to be submitted to ACS Energy Letters, July 2026.*
- **Jaeyoung Cho**, Nikhil Rampal, Stephen E. Weitzner, Graeme Henkelman, Liwen F. Wan, “Investigation of Phase Transformation Mechanisms in BN from Large-Scale Machine-Learning Molecular Dynamics,” *In preparation; Expected to be submitted to ACS Materials Letters, August 2026.*
- **Jaeyoung Cho**, Ieuan D. Seymour, Graeme Henkelman, “Revisiting the Meyer–Neldel Rule for Solid-State Materials: From Simple Metals to Solid-State Electrolytes,” *In preparation; Expected to be submitted to Journal of Materials Chemistry A, September 2026.*

PRESENTATIONS

Conference Presentations

- **Jaeyoung Cho**, Ieuan D. Seymour, Graeme A. Henkelman, “Unravelling the Diffusion Mechanism in Halide-Type Solid-State Electrolytes Beyond the Nanosecond Timescale: Adaptive Kinetic Monte Carlo with Machine-Learning Potentials,” Materials Research Society (MRS) Fall Meeting, November 30 – December 5, 2025, Boston, MA.

Invited & Seminar Talks

- “Unveiling Phase Transformation Mechanisms in BN from Large-Scale Machine-Learning MD,” Physical and Life Sciences (PLS) Division Summer Student Virtual SLAM, Lawrence Livermore National Laboratory, July 2024, Livermore, CA (virtual).
- “Long-Timescale Simulation for Unveiling the Diffusion Mechanism of Solid-State Electrolytes,” CCMS Summer Student Seminar, Lawrence Livermore National Laboratory, July 2024, Livermore, CA (virtual).

TEACHING EXPERIENCE

Teaching Assistant, Physical Chemistry Laboratory (CH 153K)

Fall 2021, Spring 2022

University of Texas at Austin

Teaching Assistant, Quantum Chemistry and Spectroscopy (CH 354)

Fall 2023, Fall 2025

University of Texas at Austin

TECHNICAL SKILLS

Methods: Ab initio calculations (DFT); molecular dynamics; transition-state search; adaptive kinetic Monte Carlo; parallel replica dynamics; metadynamics; umbrella sampling; machine-learning interatomic potentials (MACE).

Simulation Packages: VASP, CP2K, LAMMPS, EON.

Libraries & Tools: TRAVIS, VTST-Tools, Pymatgen, ASE, NumPy, SciPy, Pandas, Matplotlib, Scikit-Learn, PyTorch.

Programming: Python, Bash.